



Fig. 2: Evolution of outsourcing

lenges that impact an SME, and how outsourcing solves those are:

Limited R&D. Inadequate R&D fails to address the rapidly-changing technology and customer preferences. Most SMEs tend to use their R&D teams for engineering support. Since these teams are invariably busy solving issues with the current products, they do not have time to look at emerging trends. At times, many companies believe that the final product, when produced in volume, requires the same amount of R&D as a working prototype. They also don't include the cost of support in the cost of production. This leads to problems in the field.

Supply chain focused on cost and not on quality and availability. Most SMEs view the supply chain as a simple purchase activity and do not have a robust supply chain strategy. Mostly it is cost driven, and designers, while choosing parts, do not take into consideration aspects that impact the manufacturing process. This creates problems at the end of the product manufacturing process. Sometimes, due to short lead times, they use substandard quality products. The SMT demands selection of the right parts and product packages as these impact assembly.

Inadequate focus on product testing. One of the weakest areas, product testing has three critical elements that require focus.

1. PCBA (printed circuit board assembly) testing; also known as in-circuit testing
 2. System testing (full product testing)
 3. System calibration
- Due to volume or cost constraints,

many SMEs skip the PCBA testing. So, any problem in the board assembly is likely to be caught only in the next stage of testing. They fail to understand that when a product is tested as a whole, access to probe the parts is restricted and adequate testing cannot be carried out.

For example, if there is a fault in the PCBA in one section, which is not detected while testing the final product, a defective product is sent into the market, leading to unpredictable after-effects. This leaves SMEs with two choices—to either continue with limited testing and risk failure in the market, or identify a test method (during the design phase) and implement a test strategy that is not volume dependent but ensures the PCB is tested fully. The second approach is easy to implement when the product has a microcontroller or microprocessor, making DFT the key.

Using nonstandard manufacturing processes. Whether the board assembly is done in-house or outsourced, proper care is never given to the manufacturing process. Another area of ignorance which leads to product reliability problems is where cables and harnesses are used—products using leaded (plated through hole) parts are soldered manually. The SMT forces the assembly process to be automated and precise. It has enabled the soldering process to be lead-free (RoHS—Restriction of Hazardous Substance) and the need for control and precision has increased. With the lead-free soldering process, SMT parts have become very temperature sensitive as a lead-free part (RoHS) solders at a higher temperature, exposing components to the risk of over-heating during soldering. An uncontrolled process is detrimental to component life. SMEs cannot just handover the PCB and other components to the assembler, who solders these and gives it back. This affects the

quality of the PCBA, as well as reliability of the components.

Compliance to environmental standards. In the last decade, the focus on environment has been observed—reduction of e-waste and use of hazardous materials in the product manufacturing process through global standards like WEEE (Waste Electronics and Electrical Equipment directive), RoHS (Restriction of Hazardous Substance), REACH (Regulation on Registration, Evaluation, Authorisation and Restriction of Chemicals), etc. These standards ensure that products that come into the market do not adversely impact the environment. While these standards do not mandate conformance for existing products, they are mandatory for the new products. The biggest challenge in complying with these standards is meeting the call for changes to be incorporated in the design phase itself (i.e., if the product needs to have recyclable plastic and if the old design has metal bosses for mounting, the design has to be redone to either have an all-plastic design or the metal's bosses have to be designed as removable so that the plastic part can be recycled). This calls for focussed R&D and knowledge to handle complex designs.

Another important aspect is the harmonisation of Indian standards with global standards. Until recently, Indian standards were slack, especially in implementing standards like FCC, CE, FDA, etc. With the awareness about global standards and the associated advantage that users get from it, Indian government is forced to match up to global standards. This is making SMEs upgrade their products even in the domestic market. Building skills of this scale is not feasible and need to be outsourced.

Managing complete product engineering life cycle. Most SMEs view product engineering life cycle management as an activity-based tool. Product life cycle management (PLM) tools mainly help manage, archive and transfer product-related data in a structured manner. For example, when field-failure data is captured, it invariably brings about change and is typically carried out using a process called ECM (Engineering Change Management).